PRATIK RAJESH SAMPAT

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Education

University of Illinois at Urbana-Champaign **Aug 2022 - May 2024**
Master of Science in Computer Science

PES University **Aug 2015 - May 2019**
Bachelor of Technology in Computer Science and Engineering

Experience

AMD

Member of Technical Staff **May 2024 - Present**

- Linux Kernel Engineer for the AMD Secure Encrypted Virtualization (SEV) subsystem
- Built virtual memory hotplug to validate, accept pages on the fly in secure environments (SEV-SNP)
- Built a Symmetric Multi-Threading protection feature to defend against side-channel attacks from other guests as well as the host

University of Illinois Urbana-Champaign

Research Assistant **May 2023 - May 2024**

- Worked on bridging the fundamental disconnect between CPU bandwidth entitlement interfaces of Cloud (millicore) and OS (quota, period) leading to poor CPU allocation and autoscaling decisions
- Built an OS level autoscaler that extends the Linux bandwidth scheduler to accurately track application behavior and recommend ideal CPU limits on the fly. Published and presented paper titled: CPU Autoscaling with a kernel of truth at APSys South Korea, Oct 2025

IBM

Software Engineer - Linux Technology Center **Aug 2019 - Jul 2022**

- Device bring-up of energy and CPU features for the IBM POWER 10 Server platform
- Discovered inefficiencies in Linux CPU-idle state selection and proposed a weighted idle governor at the OS-directed Power Management Summit (OSPM) 2020. Covered by the Linux Weekly News
- Spearheaded the CPU namespace feature to virtualize and isolate topology information for containerized applications. Presented developments at the Linux Conference Australia (LCA) 2022

Software Engineering Intern **Jan 2019 - Jun 2019**

- Enabled the IBM POWER architecture on the gem5 open-source simulator
- Assisted in bringing-up and booting a full system multi-threaded Linux kernel on the simulator. Work presented in the OpenPOWER summit North America 2019

Carnegie Mellon University

Summer Undergraduate Intern **Jun 2018 - Jul 2018**

- Profiled the Operating System to extract the kernel view of memory and analyze access patterns
- Simulated a memory prefetching algorithm based on N-grams and evaluated its performance with the state of art in a simulated environment

Honor and Awards

- Patent: Haptic Feedback Glove System. Grant No. 573449 **Nov 2025**
- Full ride scholarship - University of Illinois Urbana-Champaign **Aug 2022**
- Rookie of the year - APAC region, General Manager awards, IBM **Sep 2021**
- Prof. CNR Rao Merit Scholarship, PES University **Feb 2019**
- First runner up - Honeywell Power of connected Hackathon **Jun 2017**